EESTech Challenge LC Munich 2026

Eric Schulze

Introduction (myself)

Introduction (my

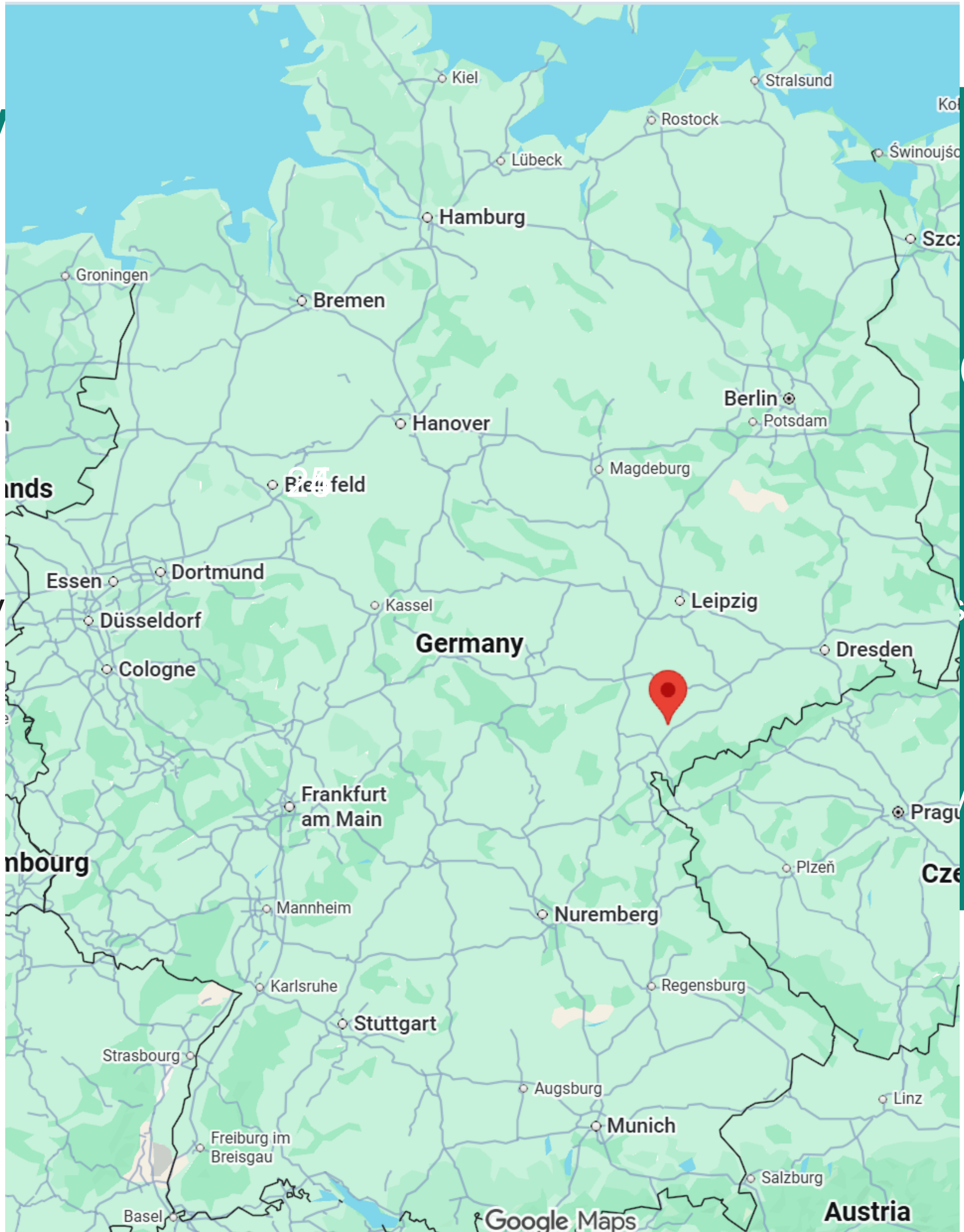
Who am I?

Where I am from?

How old am I?

Which hobbies do I hav

What is my job?



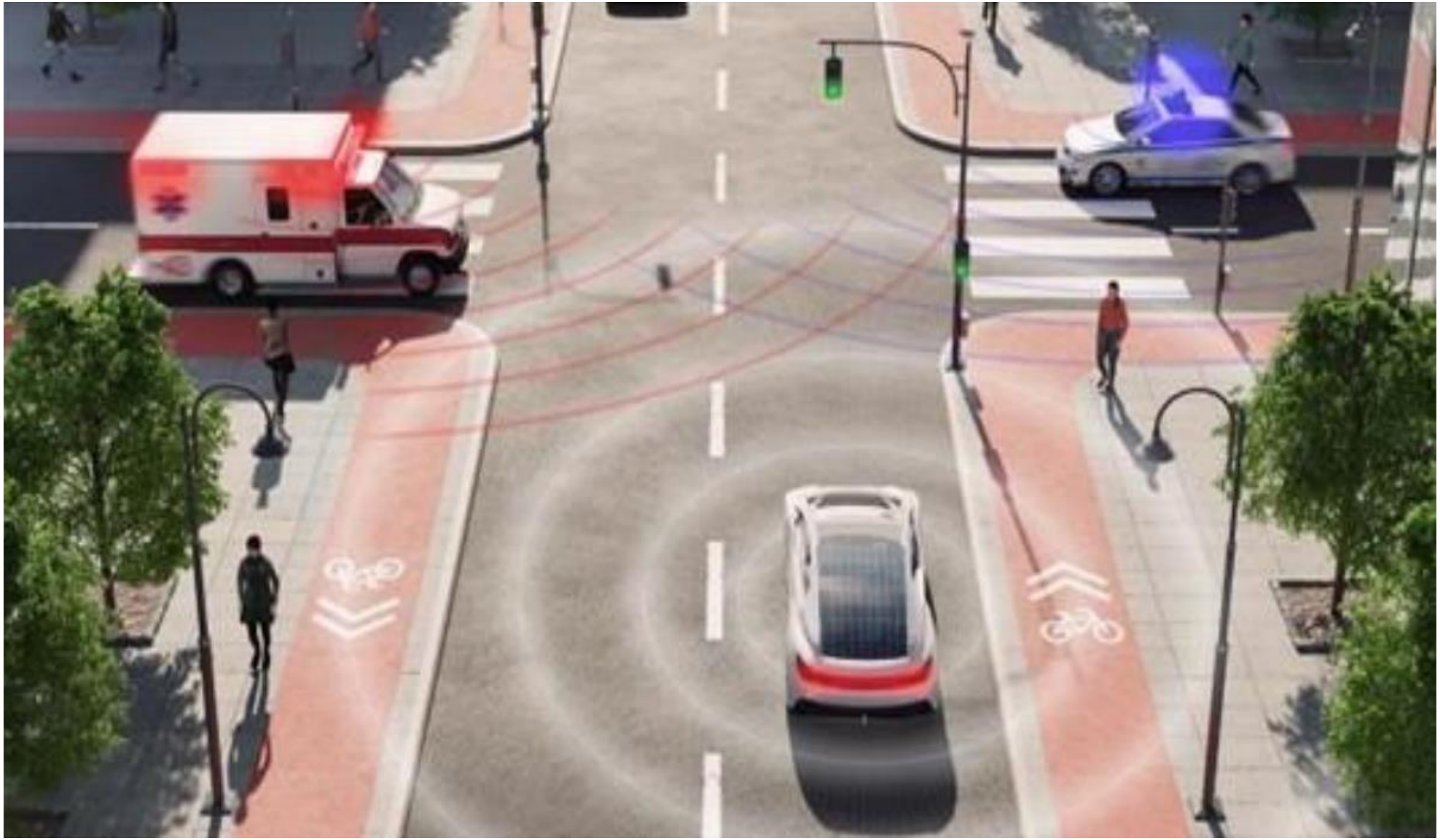
(Saxony)

, hiking

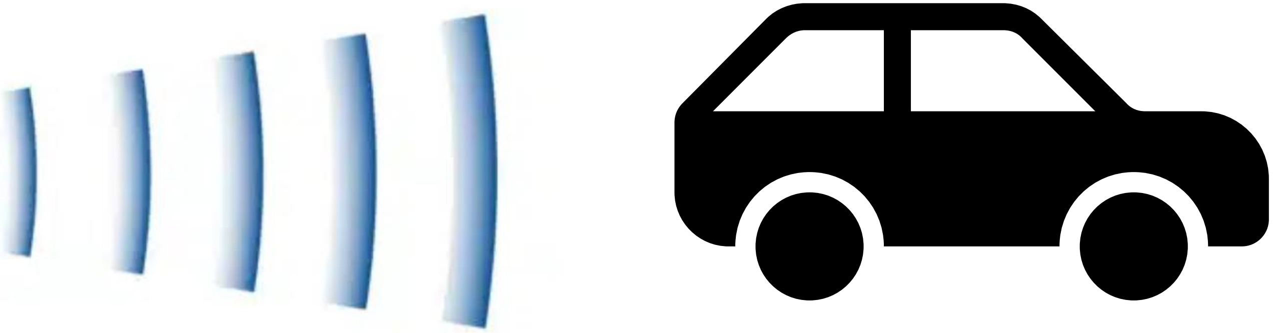
, Hackaburg)

Introduction Challenge

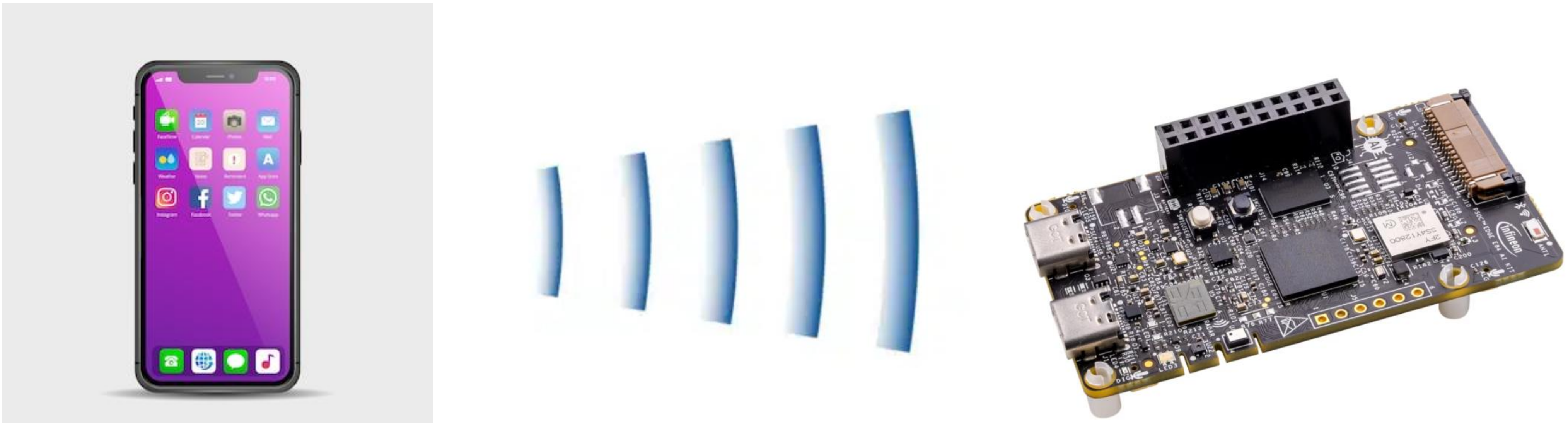
Autonomous Driving & Secure Connected Vehicles

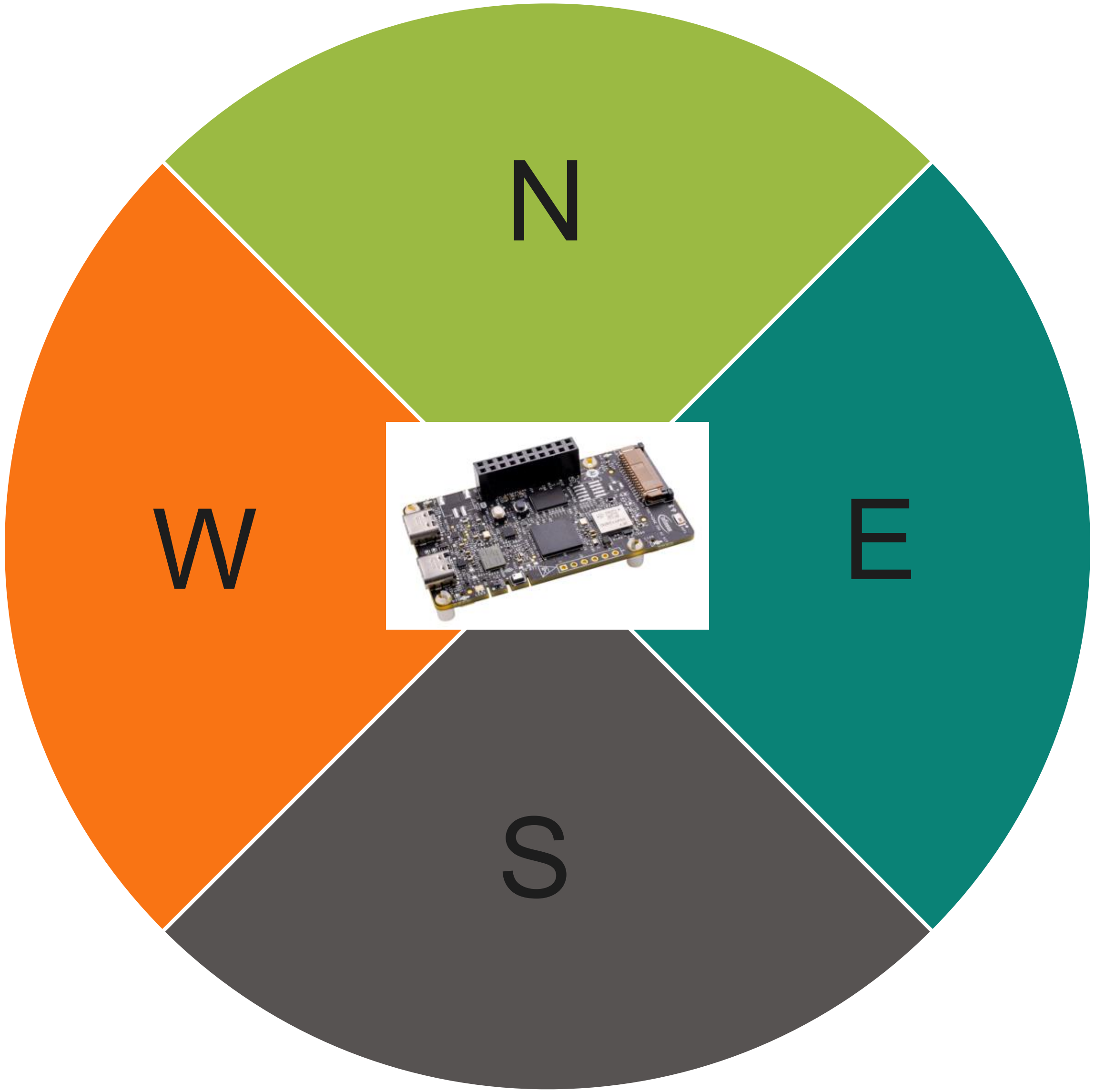


Introduction Challenge



Introduction Challenge

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Introduction Challenge

- Using the PSOC™ Edge AI Kit, your goal is to build a system that can recognize from which direction a siren or emergency signal is coming. The more accurately and reliably different directions are detected, the better your solution will perform.

- Your program / with AI model has to be flashed on the PSOC Edge AI Kit. ☺

- You can use every firetruck sound that you like. Testing your program will also be done with your device and sound that you have trained your model.

Introduction Challenge

- Develop an AI model that determines the direction of an emergency vehicle which runs on the PSOC edge
- Achieve high accuracy across multiple directional inputs.
- Test and Flash it on embedded hardware (PSOC™ Edge E84 AI Evaluation Kit)
- The code and model must be flashed on the PSOC™ Edge E84 AI Evaluation Kit 06.06 at 20:30
  (reason this will ensure that we be in the timeline on 08.06)

Evaluation scheme

Criteria	Comment	Weight
Task 1	Task 1 - Develop an AI model that determines the direction of an emergency vehicle which runs on the PSOC edge AI Kit Goal is to have many as possible direction and to be accurate as possible. Score= $\frac{1}{n}\sum_{i=1}^{n}\frac{r_i}{5} \times (1 - e^{-0.15n})$	55%
Innovation	Be creative, try to make this challenge easy as possible for you. This can be anything For example, automate the process,	20 %
Documentation	Write a short documentation of what you did and why. document also the innovative part there. ☺ Send to us the documentation and code via E-Mail or create your own github repo and send us the link (and must be send to us 07.06.2026)	20%
Feedback	Please provide feedback to us. You can write the feedback on github or send it to us via e-mail (ericjoerg.schulze@infineon.com)	5%

$$\text{Score} = \frac{1}{n}\sum_{i=1}^{n}\frac{r_i}{5} \times \left(1 - e^{-0.15n}\right)$$

- n = how many directions
- Ri = how many of 5 trys are correct (for each direction are 5 tests)

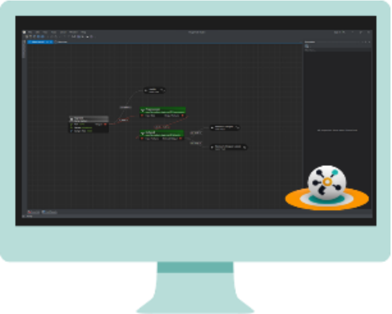
Introduction Deepcraft Studio and Modustoolbox

DEEPCRAFT™ AI Suite is your fast-track to intelligent, low-power Edge AI products

DEEPCRAFT™ products provide **innovative, trustworthy, convenient and green AI** software solutions for customers. Our **portfolio** offers **end-to-end Edge AI** and **best-in-class system performance in combination with Infineon Hardware**.

Platform & Tools

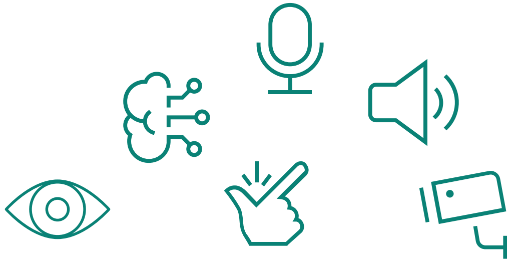
For data collection & pre-processing, model training, model conversion & deployment.



DEEPCRAFT™ Studio
20+ DEEPCRAFT™ Studio Accelerators
DEEPCRAFT™ Model Converter

Solutions

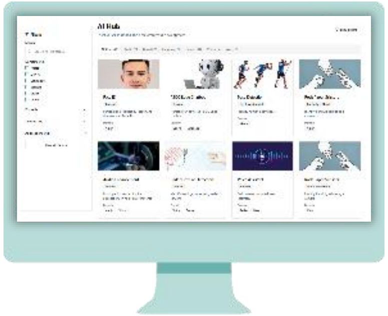
Collection of high value tools, models and configurators to add AI-based capabilities. Range across sensor and data types, from off-the-shelf to customizable solutions.



DEEPCRAFT™ Ready Models
DEEPCRAFT™ Audio Enhancement
DEEPCRAFT™ Voice Assistant

DEEPCRAFT™ AI Hub

Find the DEEPCRAFT™ Studio platform, plus the models, demos, solutions, tools and more that you need all in one centralized hub.

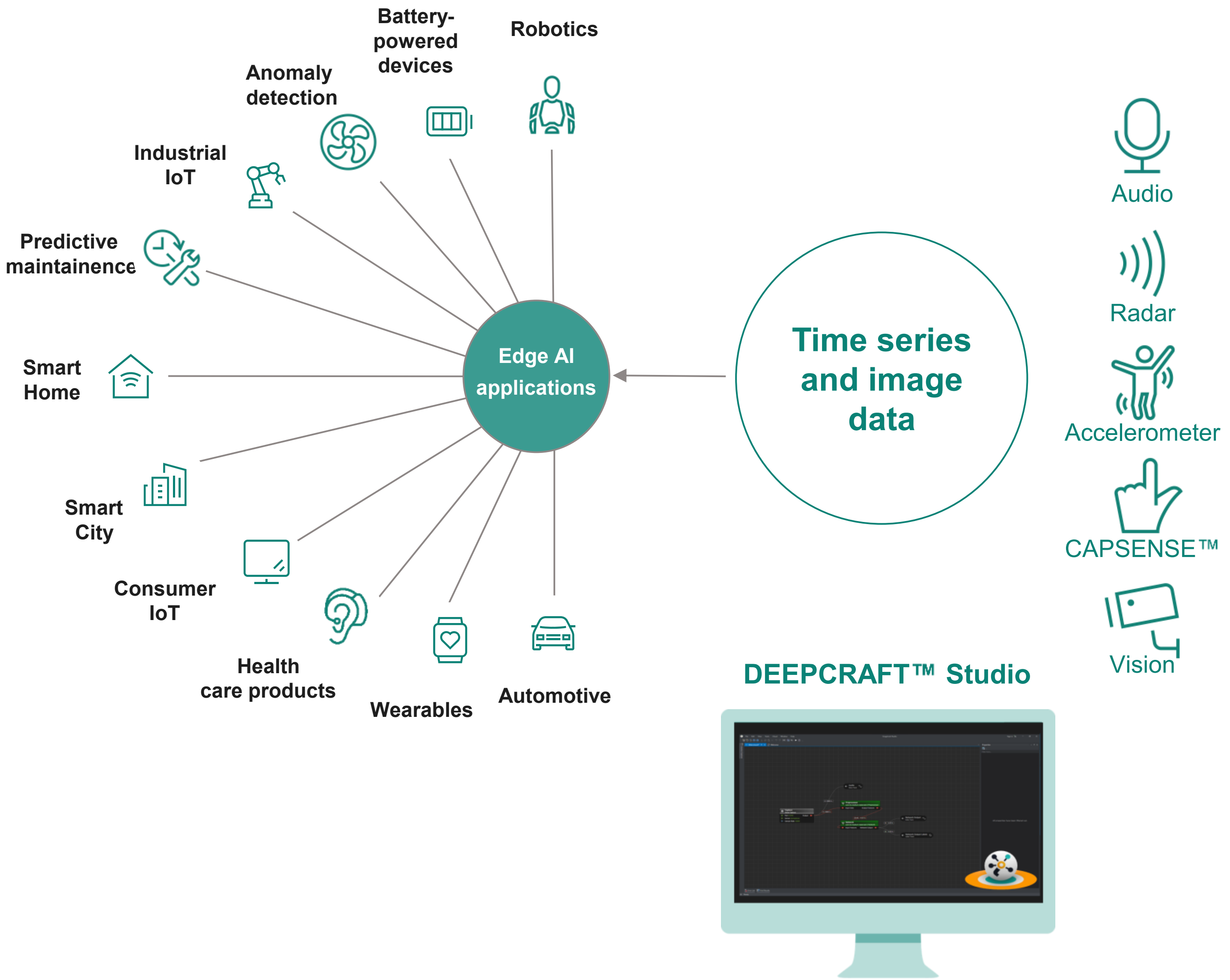


Explore the DEEPCRAFT™ AI Suite, open-source models, and more. Everything Edge AI in one place!

DEEPCRAFT™ Studio is an Edge AI development platform

DEEPCRAFT™ Studio

– Train **production-grade custom Edge AI models** with our state-of-the-art platform
– Designed for machine learning (ML) **experts and beginners**
– Leverage pre-trained Studio Accelerators for **shorter time to production**
– Transparent data policy **protects your IP**
– Graph UX interface provides **unique insight** during ML development process
– Focuses on **deep learning**, leveraging modern neural processing units for **enhanced and optimized AI capabilities**



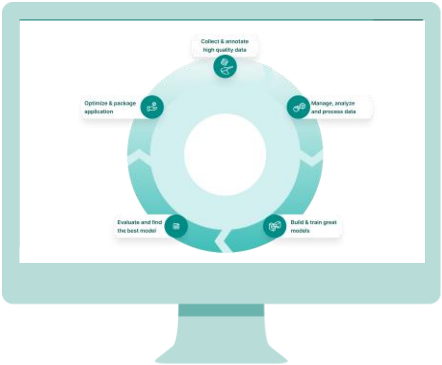
DEEPCRAFT™ Studio provides the full journey from AI/ML model development to embedded software

Build a new custom model, starting from scratch or using a DEEPCRAFT™ Studio Accelerator

Model development and customization

DEEPCRAFT™ Studio

Choose a Studio Accelerator or create a new project → Data Acquisition → Preprocessing → Model Training



Model optimization

DEEPCRAFT™ Studio

Model evaluation

Optimization

Model deployment on MCU

ModusToolbox™
- for PSOC™ and TRAVEO™

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Deployment

AURIX™ Development Studio

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Deployment

Installation and usefull tips

Installation and useful tips

- Install **Deepcraft Studio** and **Eclipse Modustoolbox** with the **Modustoolbox Setup** (and the packages that are preselected)
- If you are creating a folder. Please don‘t have a very long path, or spaces in the name (This will save you a lot of time)
- You have to create an MyInfineon account

- And you can try to import the example that we provided you on github. Try to build it.
- If it is failing, just ask us, we try to help you.

- To collect data in deepcraft Studio you have to flash a „special“ code on the board. (https://osts.infineon.com/devkit/ai-mcu-applications?id=5)

- Xensiv Radar → Gesture detection → + → select your board → connect → Experience Application with DEEPCRAFT™ Studio→ continue

Installation and useful tips

- You can watch YouTube videos from Deepcraft Studio
- 12 hours is not a lot for this challenge, so use the time efficiently as possible
- We have the example which I showed online on github (There are 2 microphones already available)

Let‘s Get Started!

github.com/Infineon/hackathon

Infineon